

① $\frac{60}{0.6m}$ cm cube placed in $E = \overset{x}{5}\overset{y}{i} - \overset{z}{4}\overset{x}{j} + \overset{z}{3}\overset{x}{k}$

The electric flux through

$A = L^2 = 0.6^2 = 0.36$

(a) $\overset{x}{y}z$ plane: $\phi = 5 \times 0.36 = 1.8$

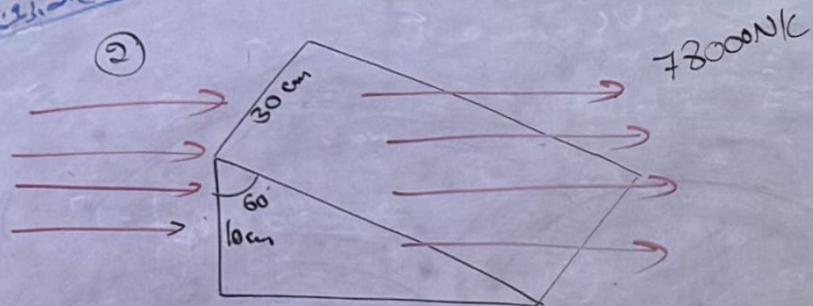
(b) $\overset{y}{x}z$ plane: $\phi = -4 \times 0.36 = ?$

(c) $\overset{z}{xy}$ plane $\rightarrow$ $\overset{x}{yz}$ plane

$\phi = 3 \times 0.36 + 5 \times 0.36$
 $= (3+5) 0.36 = ?$

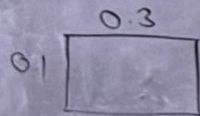
سوال نمبر 2

2



Find the electric flux through

a) Vertical area



$$A = 0.1 \times 0.3 = 0.03$$

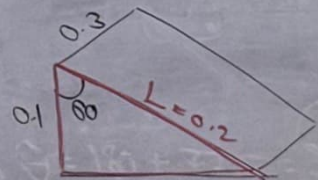
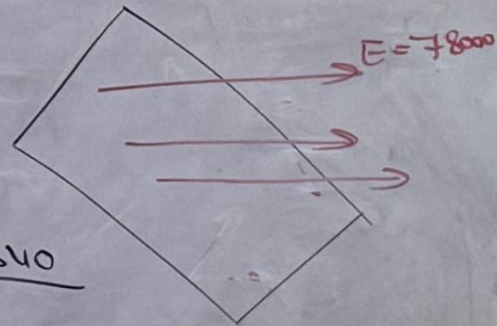
$$\Phi = -EA = -78000 \times 0.03 = -2340$$

b) Slanted area

$$A = 0.3 \times 0.2 = 0.06$$

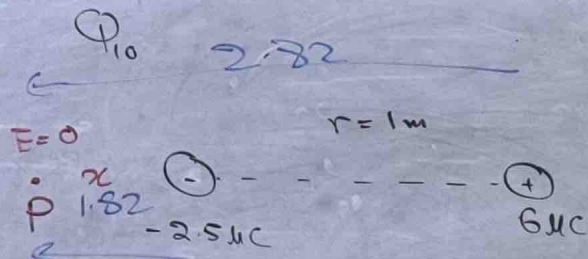
$$\Phi = EA \cos \theta$$

$$= 78000 \times 0.06 \times \cos 60 = 2340$$



$$\cos 60 = \frac{0.1}{L} \rightarrow L = \frac{0.1}{\cos 60} = 0.2$$

اسم المذنب

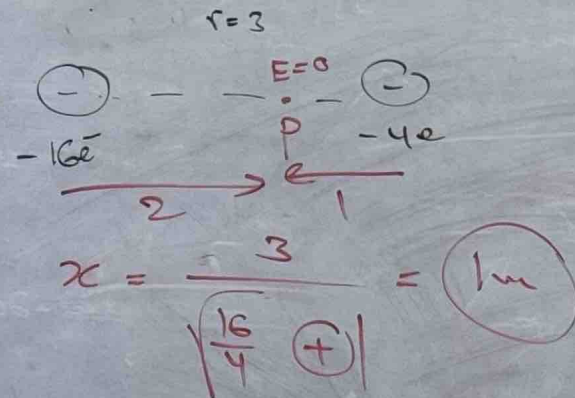


The point where $E=0$

$$x = \frac{r}{\sqrt{\frac{Q_{big}}{Q_{small}}} - 1} = \frac{1}{\sqrt{\frac{5}{2.5}} - 1} = 1.82$$

1.82m left of -2.82

2.82m left of $5\mu C$



1m left $-4e$

2m right of $-16e$